

## 电路作业第八章

8-4 1) (a)  $4\cos(2t) + 3\sin(2t) = 5\cos(2t - 37^\circ)$

(b)  $-6\sin(5t - 75^\circ) = 6\cos(5t + 15^\circ)$

(2) (a)  $6 - j8 = 10 \angle \arctan(-\frac{8}{6}) = 10 \angle -53.1^\circ \rightarrow 10\cos(\omega t - 53.1^\circ)$

(b)  $-8 + j6 = 10 \angle \arctan(-\frac{6}{8}) = 10 \angle 143.1^\circ \rightarrow 10\cos(\omega t + 143.1^\circ)$

(c)  $-j10 = 10 \angle -90^\circ \rightarrow 10\cos(\omega t - 90^\circ)$

8-6. (✓) 若  $f_1(t) = 6\cos(\omega t - 72^\circ)$ ,  $f_2(t) = 12\sin(\omega t + 150^\circ)$ , 求  $f_1(t) + f_2(t)$  及  $f_1(t) - f_2(t)$  的最大值。

$f_1(t) = 6 \angle -72^\circ$      $f_2(t) = 12\cos(\omega t + 60^\circ) = 12 \angle 60^\circ$

$f_1(t) + f_2(t) = 1.854 - j5.706 + 6 + j10.392 = 7.854 + j4.686 = 9.146 \angle 30.82^\circ$

$f_1(t) - f_2(t) = 1.854 - j5.706 - 6 - j10.392 = -4.146 - j16.098 = -16.62 \angle 75.56^\circ$

∴  $f_1 + f_2$  最大值 9.146     $f_1 - f_2$  最大值 16.62

8-9 已知元件 A 的正弦电压  $u(t) = 12\cos(1000t + 30^\circ)$  V, 求流过元件 A 的正弦电流  $i(t)$ , 若 A 为: (1) 电阻, 且  $R = 4 \text{ k}\Omega$ ; (2) 电感, 且  $L = 20 \text{ mH}$ ; (3) 电容, 且  $C = 1 \text{ }\mu\text{F}$ 。绘三种情况的相量图。

$\dot{U}_m = 12 \angle 30^\circ$      $\dot{I}_{Rm} = \frac{\dot{U}_m}{R} = 0.003 \angle 30^\circ$

(1)

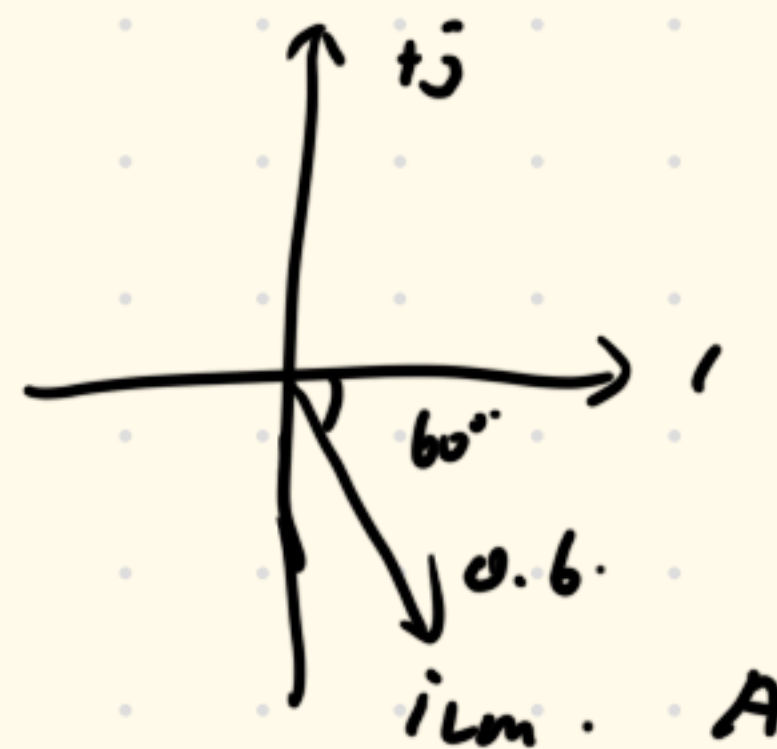
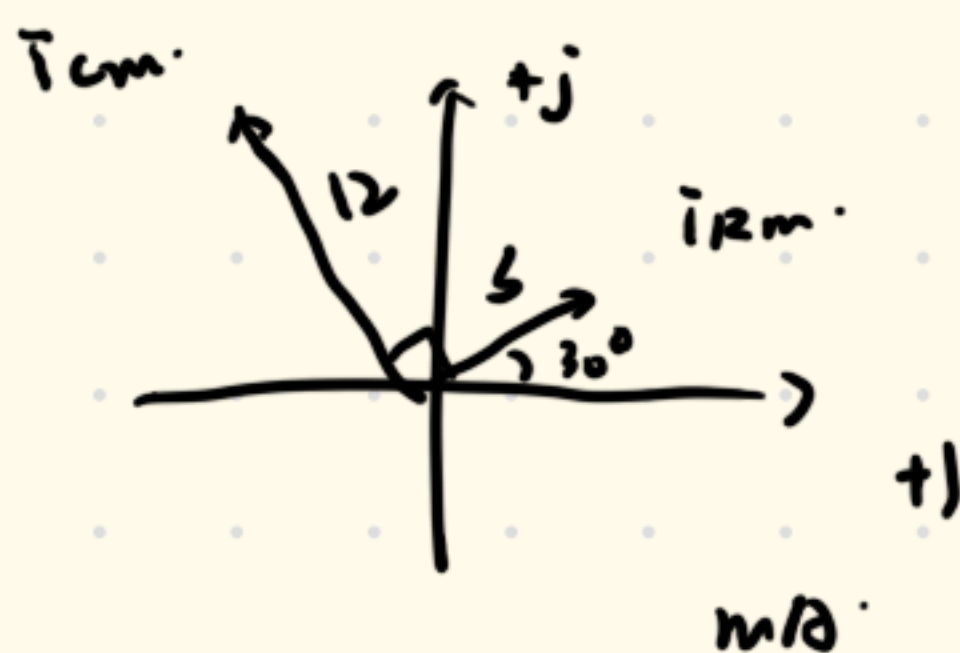
$i_R(t) = \frac{3}{1000} \cos(1000t + 30^\circ) = 0.003 \cos(1000t + 30^\circ) \text{ A}$   
 $= 3 \cos(1000t + 30^\circ) \text{ mA}$

(2)  $\dot{I}_{Lm} = \frac{\dot{U}_m}{j\omega L} = \frac{12 \angle 30^\circ}{20 \angle 90^\circ} = 0.6 \angle -60^\circ \text{ A}$

$i_L(t) = 0.6 \cos(1000t - 60^\circ) \text{ A}$

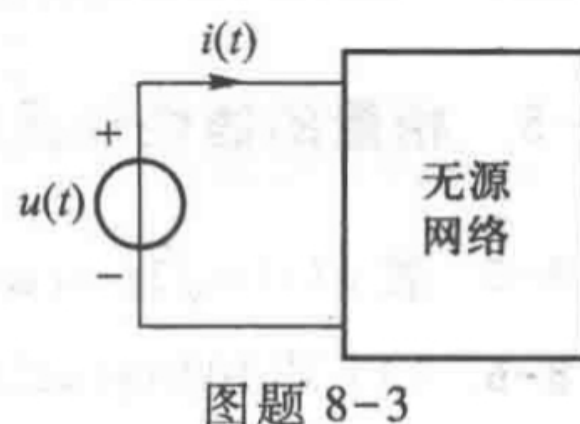
(3)  $\dot{I}_{Cm} = \frac{\dot{U}_m}{-j\frac{1}{\omega C}} = \frac{12 \angle 30^\circ}{\frac{1}{1000 \cdot 10^{-6} \angle 90^\circ}} = 12 \times 10^{-3} \angle 120^\circ$

$i_C(t) = 12 \cos(1000t + 120^\circ) \text{ mA}$



8-11 已知图题 8-3 所示无源网络两端的电压  $u(t)$  和电流  $i(t)$  各如下式所示。试求每种情况时的阻抗及导纳。

- (1)  $u(t) = 200\cos(314t) \text{ V}, i(t) = 10\cos(314t) \text{ A};$   
 (2)  $u(t) = 10\cos(10t+45^\circ) \text{ V}, i(t) = 2\cos(10t+35^\circ) \text{ A};$   
 (3)  $u(t) = 100\cos(2t+30^\circ) \text{ V}, i(t) = 5\cos(2t-60^\circ) \text{ A};$   
 (4)  $u(t) = 40\cos(100t+17^\circ) \text{ V}, i(t) = 8\cos(100t) \text{ A};$



- (5)  $u(t) = 100\cos(\pi t - 15^\circ) \text{ V}, i(t) = \sin(\pi t + 45^\circ) \text{ A};$   
 (6)  $u(t) = [-5\cos(2t) + 12\sin(2t)] \text{ V}, i(t) = 1.3\cos(2t + 40^\circ) \text{ A};$   
 (7)  $u(t) = \operatorname{Re}[je^{j2t}] \text{ V}, i(t) = \operatorname{Re}[(1+j)e^{j(2t+30^\circ)}] \text{ mA}.$

$$(3) Z = \frac{100 \angle 30^\circ}{5 \angle -60^\circ} = 20 \angle 90^\circ = j20 \Omega.$$

$$Y = \frac{1}{Z} = \frac{1}{j20} = -j0.05 \text{ S}.$$

$$(6) u(t) = -13 \cos(2t + 67.38^\circ) \text{ V} = -13 \angle 67.38^\circ \text{ V}$$

$$i(t) = 1.3 \angle 40^\circ \text{ A}.$$

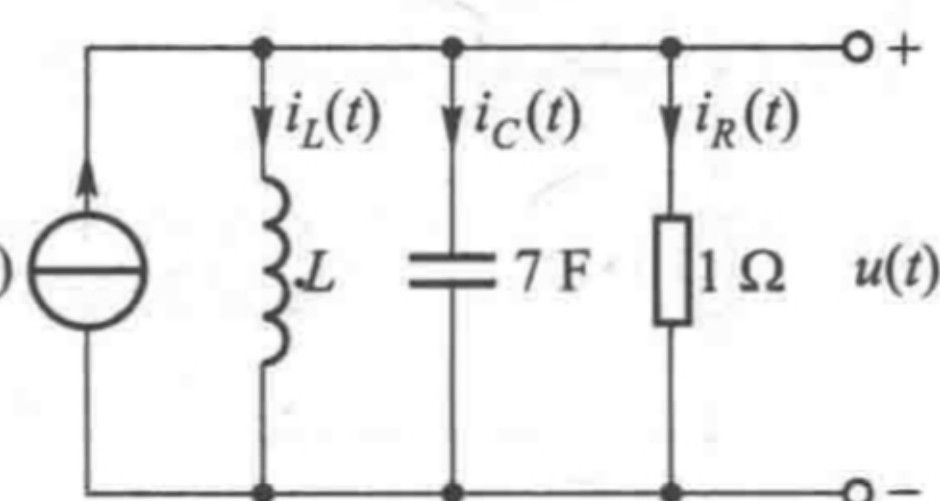
$$Z = \frac{-13 \angle 67.38^\circ}{1.3 \angle 40^\circ} = -10 \angle 27.38^\circ \Omega = 10 \angle -152.62^\circ \Omega$$

$$Y = \frac{1}{Z} = \frac{1}{10 \angle -152.62^\circ} = 0.1 \angle 152.62^\circ \text{ S}.$$

8-13 图题 8-5 所示电路中, 施加于电路的电流源电流为  $i(t) = (8\cos t - 11\sin t) \text{ A}$ , 已知电压  $u(t) = (\sin t + 2\cos t) \text{ V}$ 。求  $i_R(t)$ 、 $i_C(t)$ 、 $i_L(t)$  以及  $L$ , 并绘出相量图。

$$i(t) = 8 + j11 \text{ A} = 13.6 \angle 53.97^\circ \text{ A} = \dot{I}_m$$

$$u(t) = 1 \angle 90^\circ + 2 \angle 0^\circ \text{ V} = 2 - j1 \text{ V} = \sqrt{5} \angle -26.56^\circ \text{ V} = \dot{U}_m$$



图题 8-5

$$\dot{I}_{Rm} = \frac{\dot{U}_m}{R} = \sqrt{5} \angle -26.56^\circ \text{ A}$$

$$\therefore i_R(t) = \sqrt{5} \cos(t - 26.56^\circ) = 2.236 \cos(t - 26.56^\circ) \text{ A}.$$

$$\dot{I}_{Cm} = \frac{\dot{U}_m}{-j\frac{1}{7}} = \frac{\sqrt{5} \angle -26.56^\circ}{7 \angle 90^\circ} = 7\sqrt{5} \angle 90 - 26.56^\circ \text{ A}.$$

$$\therefore i_C(t) = 15.652 \cos(t + 63.44^\circ) \text{ A}.$$

$$\therefore \dot{I}_{Lm} = \dot{I}_m - \dot{I}_{Rm} - \dot{I}_{Cm} = 13.6 \angle 53.97^\circ - 2.236 \angle -26.56^\circ - 15.652 \angle 63.44^\circ$$

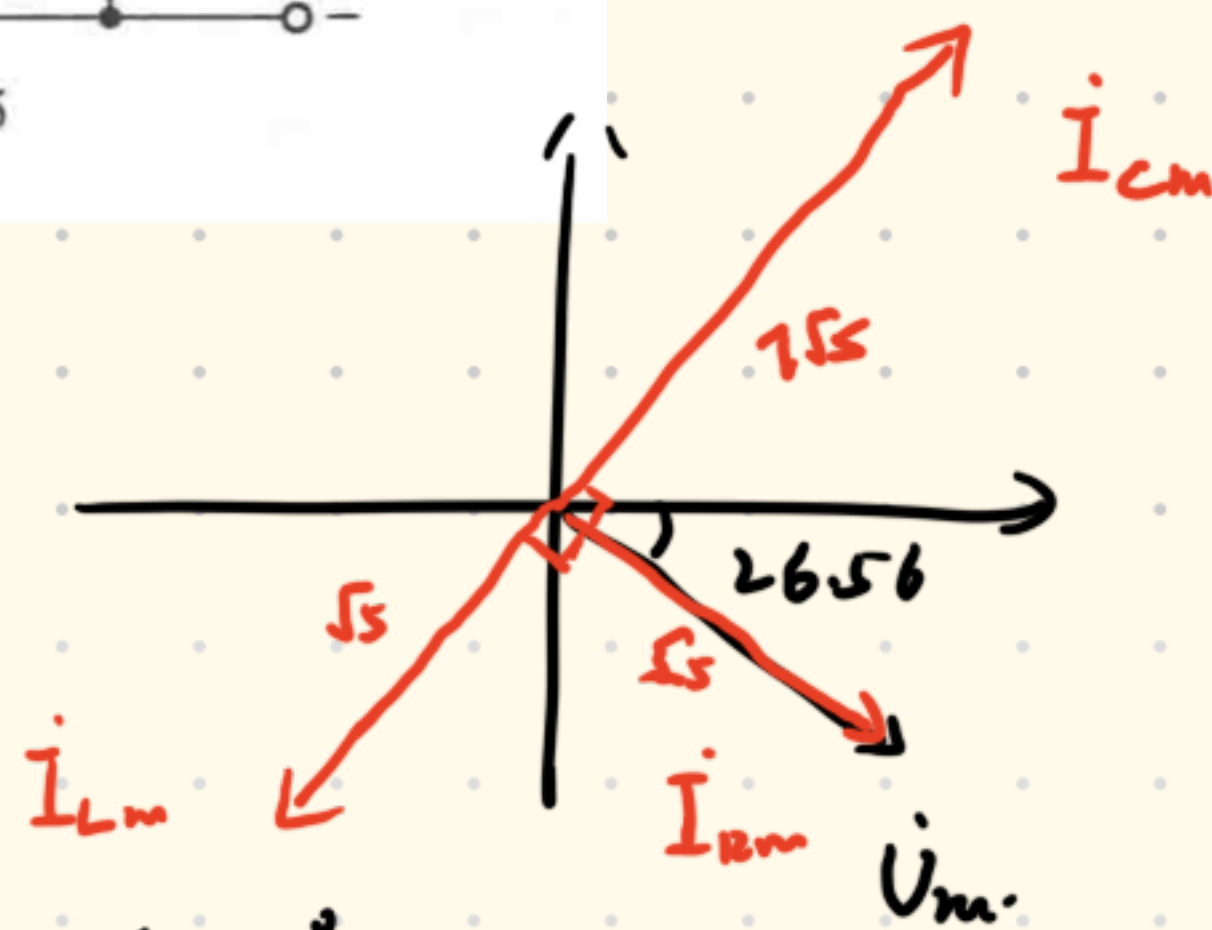
$$= 8 + j11 - (2 - j1) - (7 + j14)$$

$$= -1 - j2 = -(1 + j2) = 2.236 \angle -116.56^\circ \text{ A}.$$

$$\therefore i_L(t) = 2.236 \cos(t - 116.56^\circ) \text{ A}.$$

$$\therefore j\omega L = \frac{\dot{U}_m}{\dot{I}_{Lm}} = \frac{2 - j}{-(1 + j2)} = \frac{(2 - j)(1 - j2)}{-5} = \frac{2 - 2j - j - 2}{-5} = j$$

$$\therefore L = 1 \text{ H}$$



8-15 图题 8-7 所示电路中  $u_s(t) = U_{sm} \cos t$ , 求  $u_o(t)$ 。

8-16 电路如图题 8-8 所示, 已知  $u_c(t) = \sqrt{2} \cos(2t) \text{ V}$ , 试求电源电压  $u_s(t)$ 。绘所有电压、电流的相量图。

$$\frac{1-j\omega L}{1+j\omega L} = \frac{j}{1+j} = \frac{j(1-j)}{2} = \frac{1+j}{2} = 0.5 + j0.5$$

$$\dot{U}_{om} = \dot{U}_{sm} \cdot \frac{-j}{0.5 + j0.5 + (1-j)}$$

$$= \dot{U}_{sm} \cdot \frac{j}{-0.5 + j0.5}$$

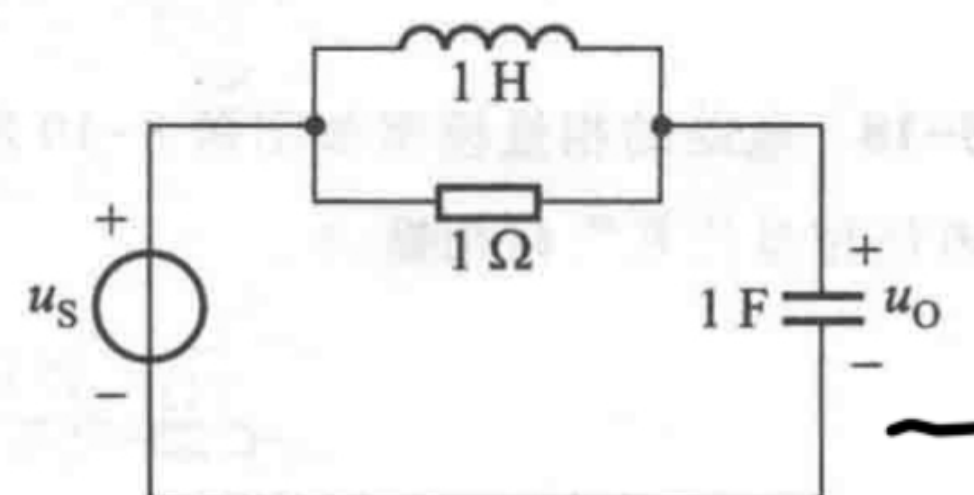
$$= \dot{U}_{sm} \cdot \frac{j(0.5 + j0.5)}{0.5^2 + 0.5^2}$$

$$= \dot{U}_{sm} \cdot (-1 + j)$$

$$= \sqrt{2} \dot{U}_{sm} \angle 0^\circ \angle -45^\circ$$

$$= \sqrt{2} \dot{U}_{sm} \angle -45^\circ$$

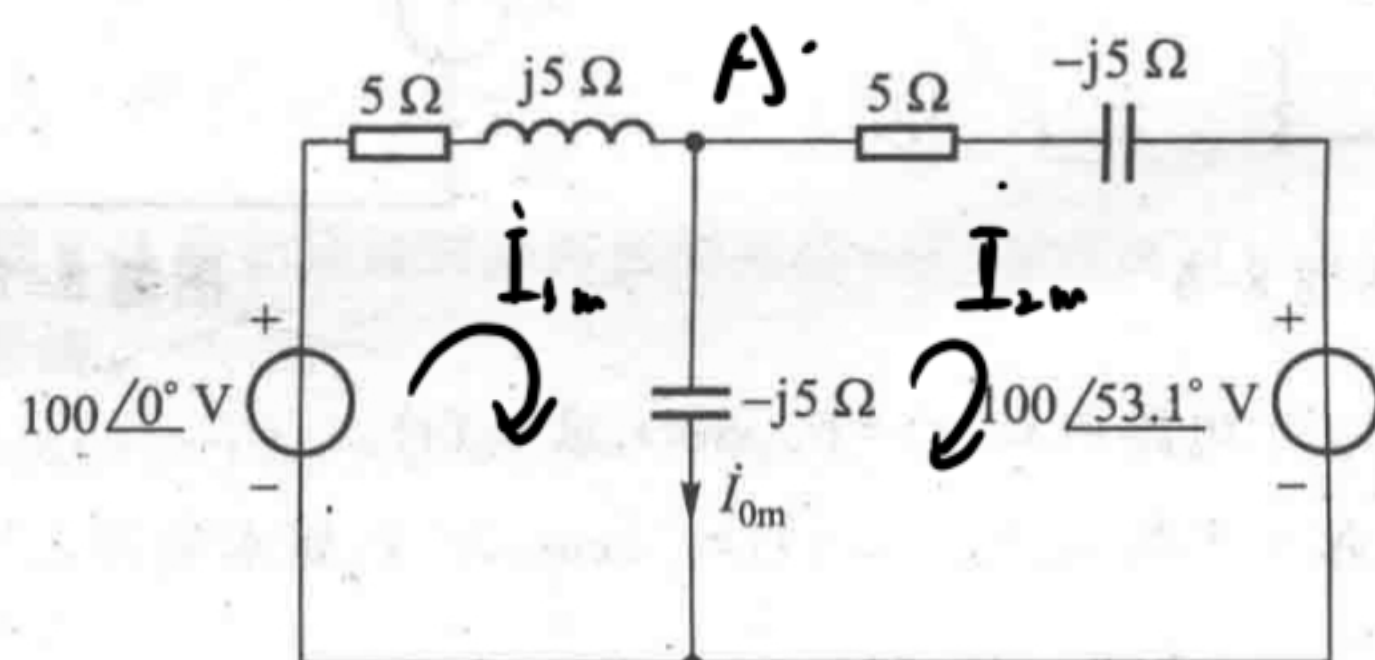
$$\therefore u_o(t) = \sqrt{2} U_{sm} \cos(t - 45^\circ) \text{ V}$$



图题 8-7

8-18 电路的相量模型如图题 8-10 所示, 试分别用网孔分析和节点分析求解  $i_{om}$ 。(各电源电压相量均系振幅相量。)

只列方程

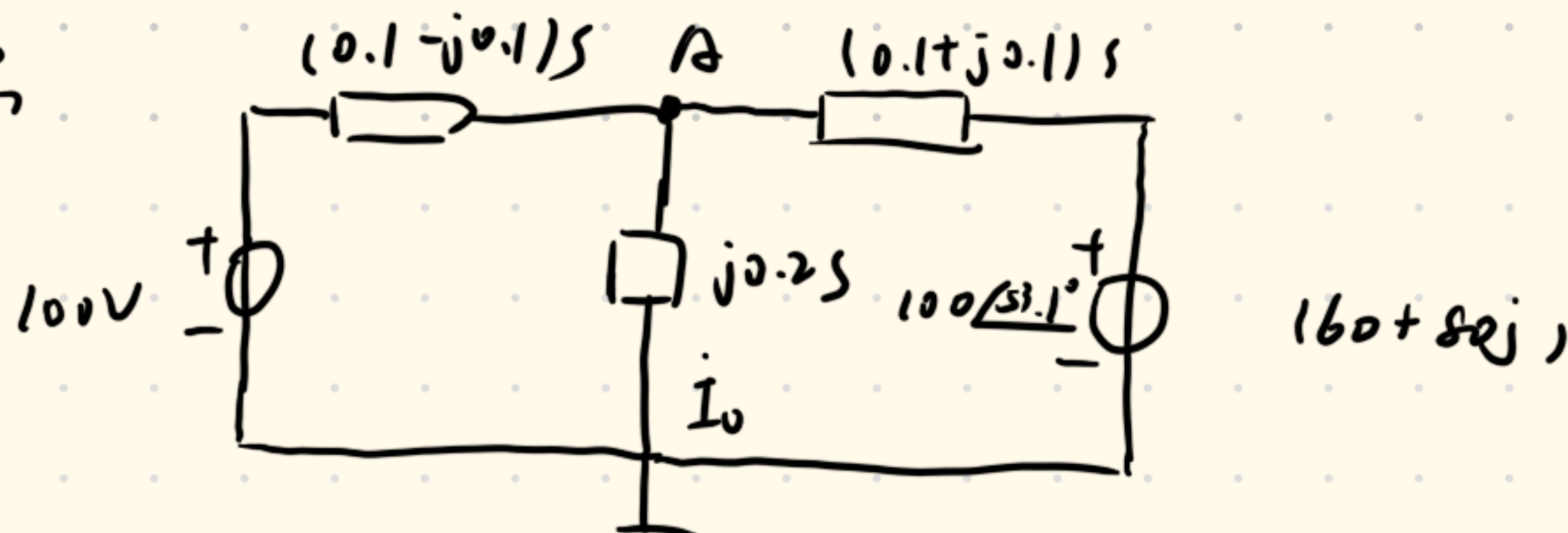


图题 8-10

$$\text{网孔: } (5 + j5 - j5) \dot{I}_{1m} - (-j5) \dot{I}_{2m} = 100 \angle 0^\circ$$

$$(5 - j5 - j5) \dot{I}_{2m} - (j5) \dot{I}_{1m} = -100 \angle 53.1^\circ$$

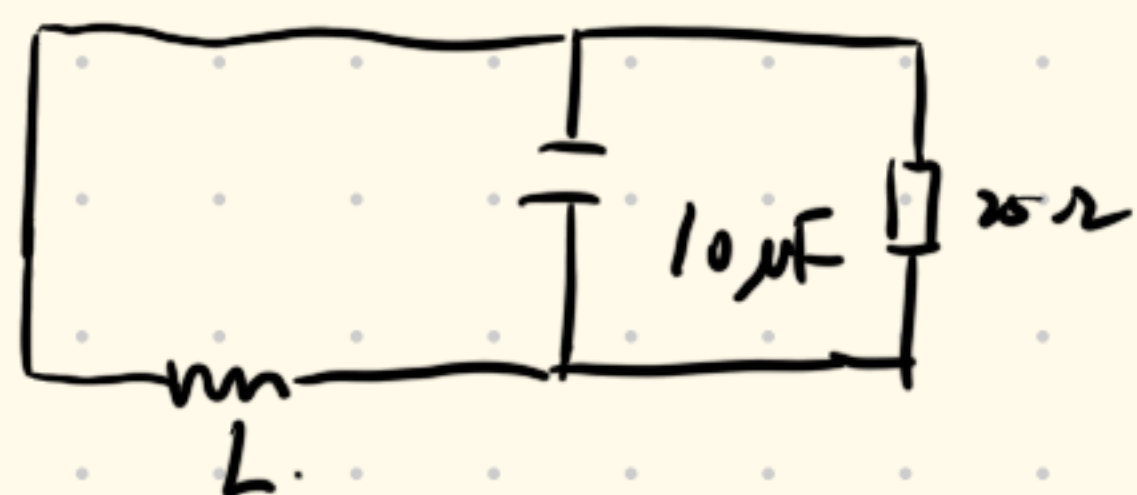
节点



$$(0.1 - j0.1 + 0.1 + j0.1 + j0.2) \dot{U}_A = 100 \angle 0^\circ (0.1 - j0.1) + 100 \angle 53.1^\circ (0.1 + j0.1)$$

$$= 100 (0.1 - j0.1) + (60 + 80j) (0.1 + j0.1)$$

8-25  $10\ \mu\text{F}$  电容与  $25\ \Omega$  电阻并联, 问当  $\omega = 8000\ \text{rad/s}$  时, 需用多大的电感与这一并联组合串联, 使整个电路的电抗分量为零?



$$8000 \times 10^{-5} = 8 \times 10^{-2} = 0.08$$

$$\frac{8}{100}$$

$$3125$$

$$31250$$

$$Z = j\omega L + \frac{25 \cdot \frac{-j}{\omega C}}{25 - j \frac{1}{\omega C}} = j8000L + \frac{25(-j125)}{25 - j125}$$

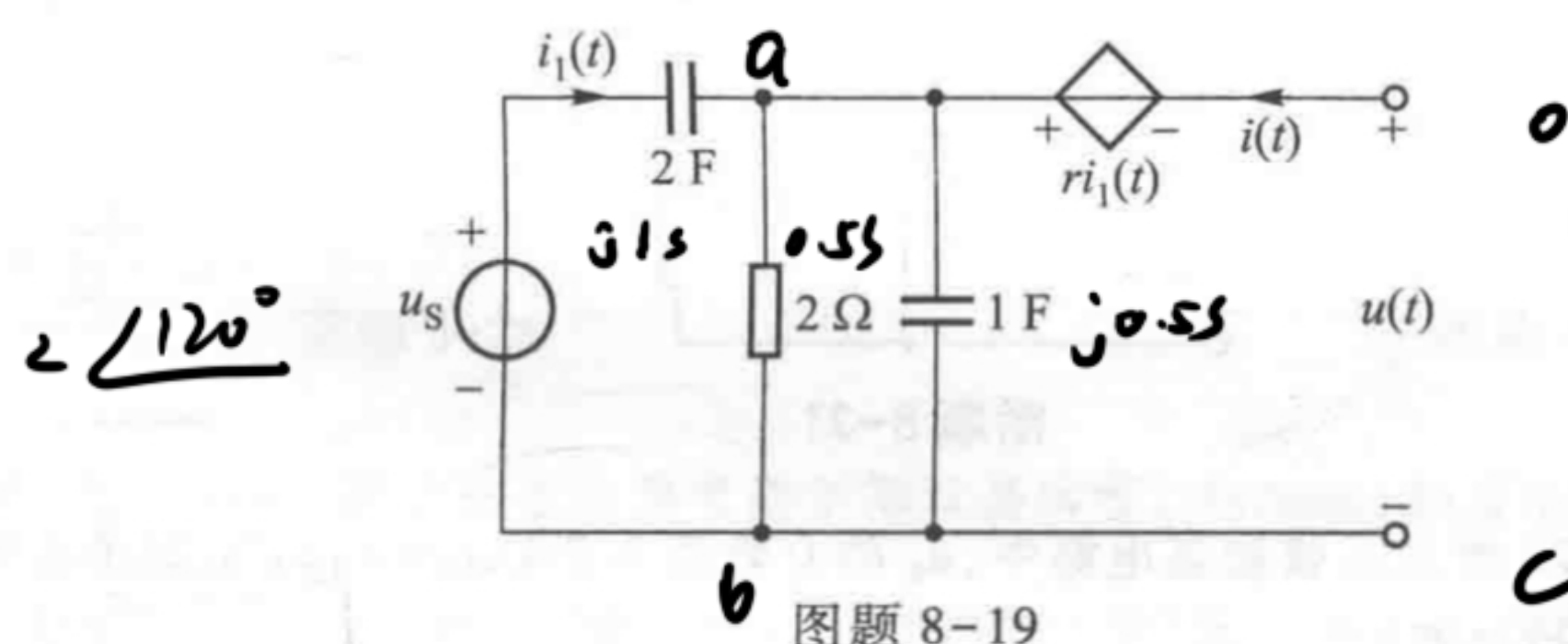
$$= j8000L - \frac{j \cdot 25^2}{50 - j25}$$

$$= j8000L - \frac{j \cdot 25^2 (50 + j25)}{50^2 + 25^2}$$

$$= 5 + j(8000L - 10)$$

$$8000L - 10 = 0 \Rightarrow L = \frac{1}{800} = 1.25\ \text{mH}$$

8-28 求图题 8-19 所示正弦稳态电路的戴维南和诺顿相量模型, 已知  $u_s(t) = 2 \cos(0.5t + 120^\circ)\ \text{V}$ ,  $r = 1\ \Omega$ 。



$$U_{ab}(j1.5 + 0.5) = jU_s \Rightarrow U_{ab} = \frac{jU_s}{0.5 + j1.5}$$

$$I_1 = j(U_s - U_{ab})$$

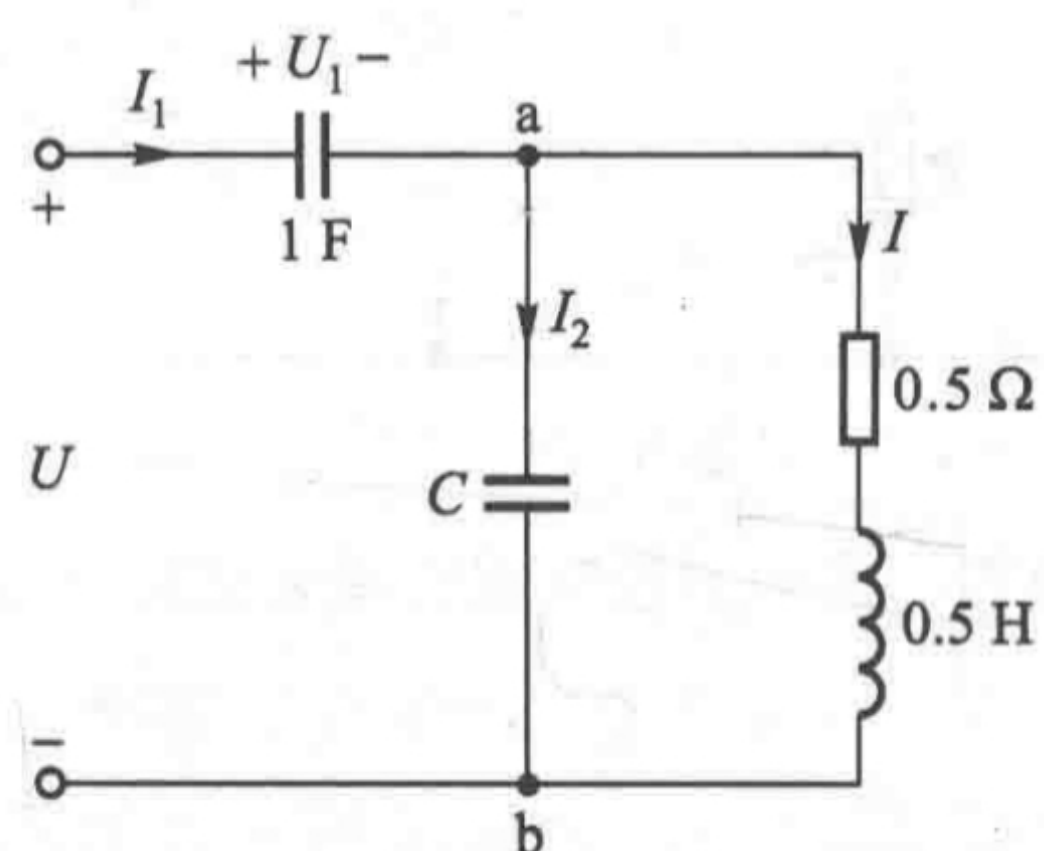
$$U_{oc} = -I_1 + U_{ab}$$

$$= -j(U_s - \frac{jU_s}{0.5 + j1.5}) + \frac{jU_s}{0.5 + j1.5}$$

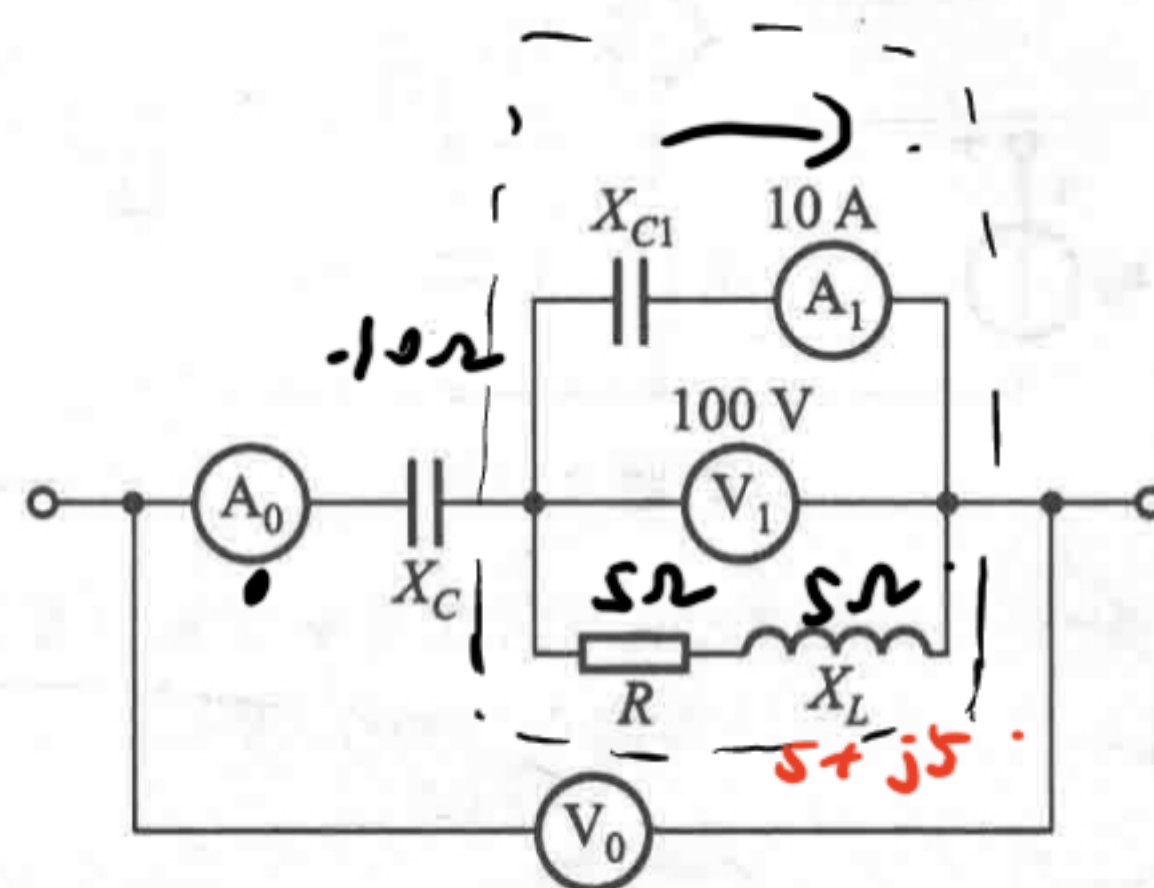
$$= \frac{-jU_s(0.5 + j1.5) + (-U_s) + jU_s}{0.5 + j1.5} = \frac{0.5U_s + j0.5U_s}{0.5 + j1.5} = U_s \frac{1 + j}{1 + j3} = 2 \angle 120^\circ \frac{1.414 \angle 45^\circ}{3.162 \angle 71.56^\circ}$$

$$= 0.8944 \angle 93.44^\circ$$

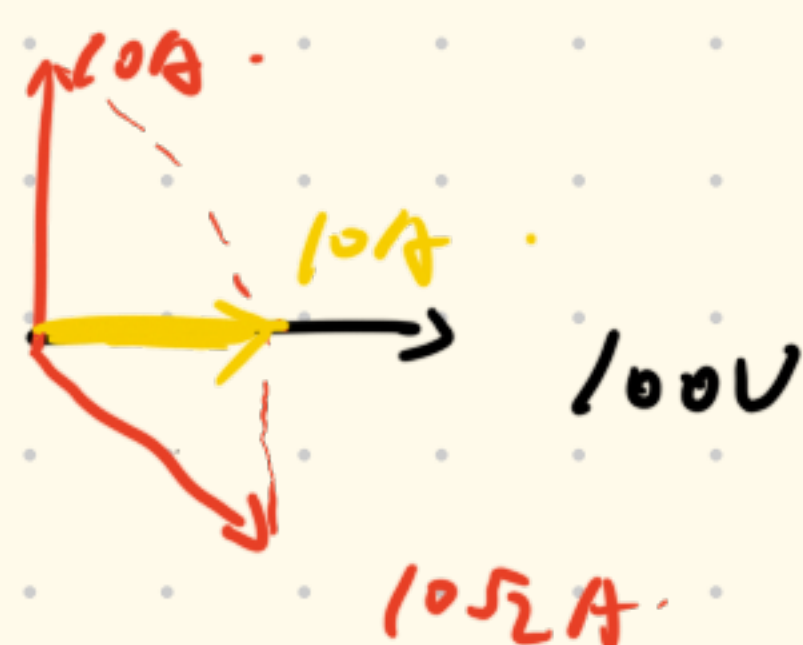
8-31 电路如图题 8-25 所示, 已知  $X_C = -10 \Omega$ 、 $R = 5 \Omega$ 、 $X_L = 5 \Omega$ , 各仪表指示有效值, 试求  $A_0$  及  $V_0$  的读数。



图题 8-24



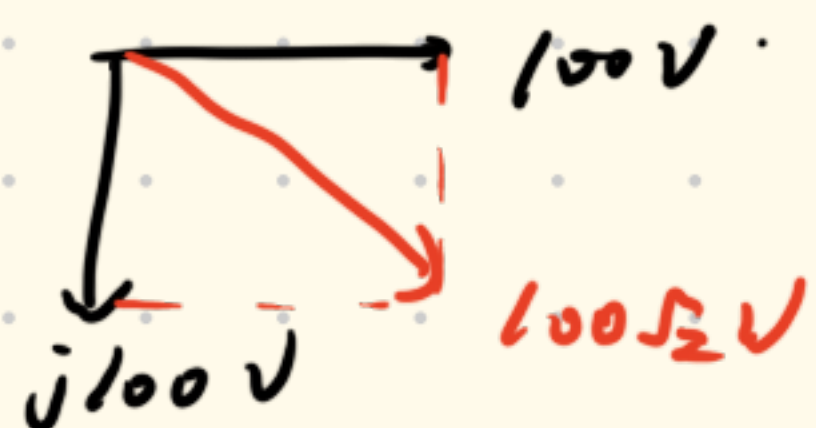
图题 8-25



$$\frac{100}{5+j5} = \frac{20}{1+j} = 10(1-j)$$

$$A_0: 10A$$

$$10 \times (-j10) = -100j V$$



$$V_0: 100.52 \approx 141.4 V$$